

E3 Classes

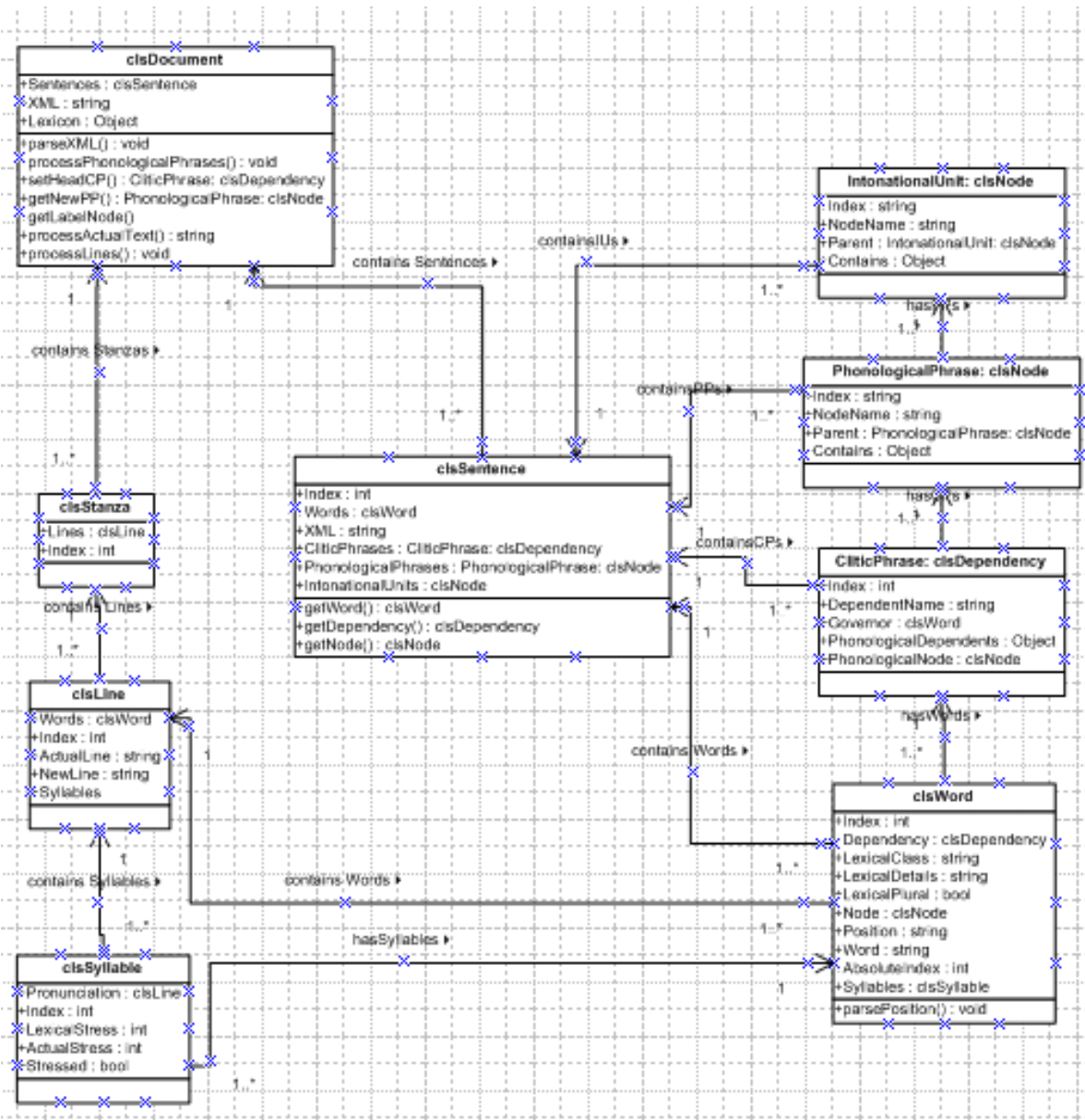


Figure 13: Phonological Classes

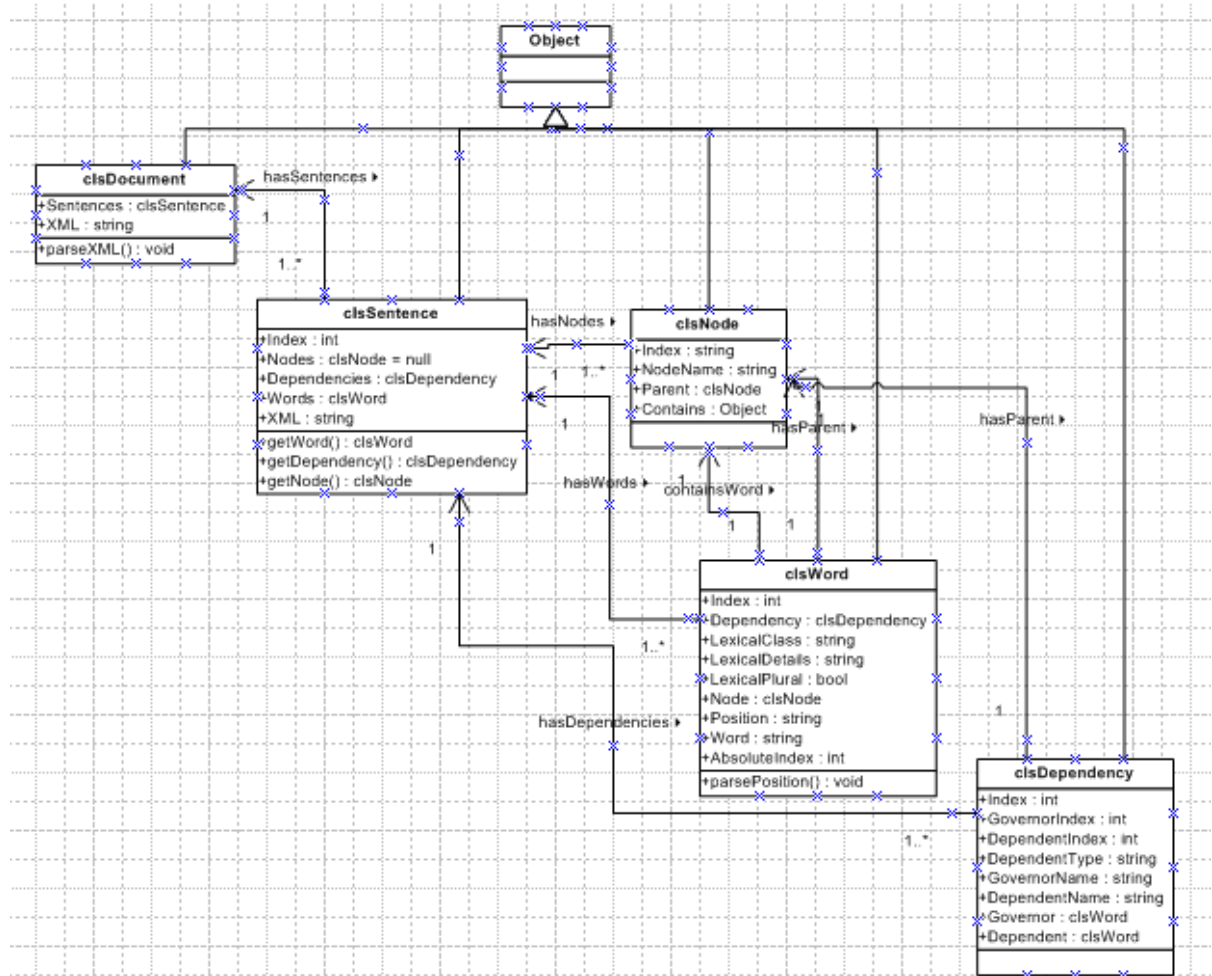


Figure 14: Syntactic Classes

E4 Stress Assignments

The program has three phases for stress assignments:

- Lexical stresses
- Phrase stresses
- Relative stresses

These are discussed in more detail below.

E4.1 Lexical Stresses

Lexical stresses are assigned to a word from its dictionary entry (derived primarily from Ward, 1997). There are three levels of stress – primary (2), secondary (1) and no stress (0). Content words are assigned a primary stress on the relevant syllable, but non-content words are given a secondary stress where a primary stress is marked in the dictionary (non-content words include prepositions and pronouns). This is to indicate the difference in stress levels between content and non-content words.

E4.2 Phrase Stresses

Phrase stresses are assigned by using the Compound Stress and Nuclear Stress Rules developed by Hayes (1984b) and others. The compound stress rule adjusts the stress in compounds which fit particular semantic categories. Stress clashes are then resolved. Finally, increasing phrase stress is assigned to the primary stressed syllables in content words as a phrase progresses.

E4.3 Relative Stresses

Relative stresses are assigned to the syllables based on their phrase stress values. Initially, syllables with no phrase stress (0) are assigned weak stress (w), syllables with secondary stress (1) are assigned low stress (n) and syllables with a phrase stress of 2 or greater are assigned strong stress (s).

These values are modified if there is a sequence of two contiguous relative stresses with the same value. The goal is to avoid contiguous stresses with the same value. However, the system still generates such sequences where there is no relationship between the syllables. The syllables are tested for their syntactic relationship: if one syllable governs the other then it is assigned stronger stress, or the governed syllable is given weaker stress (depending on the value of the syllable immediately before the sequence). If there is no syntactic relationship, then the stresses are left unadjusted. Additional considerations also affect the relative stress resolution of identical contiguous stresses: for example, prepositions will receive greater stress, except where they govern a pronoun which will receive greater stress than the preposition.

Phase	Stresses	Comment
Shall I compare thee to a summer's day?		
Lexical Stress	2102111202	Non-content words "I" and "thee to a" all receive secondary stress (1)
Phrase Stress	2103111405	Content words receive an additional stress to achieve a stress gradient across the phrase
Relative Stress	snwswnws	In the sequence "111", "to" is promoted to n stress because it governs "a"
Scandroid	wswnws	To is allocated an ambiguous stress because it is the second weak stress in a series of three (www)
If hairs be wires, black wires grow on her head		
Lexical Stress	1222,222112	
Phrase Stress	1234,567118	
Relative Stress	nsms,nmsnws	In the sequence "222,", "be" is demoted to m because the copula ("wires") of the verb to be is considered to govern the verb. In the sequence ",222", "black" is demoted to m because it is governed by "wires". "wires" is demoted to m because it is governed by "grow". "on" is promoted to n because it governs "her"
AnalysePoems	0202222002 (wswn,ssswws)	
Scandroid	wswnssswws	

Figure 15: examples of the stress assignment procedure

E4.4 Discussion

The system generates a relative stress pattern which approximates what I think are the stress values of a phrase, and which produces scansion which are closer to expert opinions than other stress assignment systems (such as AnalysePoems or Scandroid). The system fixes the stress values of the line, and adjusts the metrical pattern round them, rather than adjusting the stress values to fit metrical patterns.

In most cases, this reflects the practice of native speakers: it seems that only where certain types of polysyllabic words appear in compounds are the stresses adjusted to fit the meter.

Moreover, this relative stress assignment allows, for example, the promotion of prepositions to strong metrical positions, which is required in some verses. Plamondon achieves this by adjusting the stress of the line to fit the meter, but the relative stress assignment system uses only linguistic assumptions to produce the same effect.

E5 Output

The following information is produced by the application when it scans a line.

E5.1 Antelope Parser Syntactic output

```
<?xml version="1.0" encoding="utf-8" ?>
```

```
= <Document id="1">
= <Sentence id="1">
= <words>
  <word ind="1" pos="VBD">Shall</word>
  <word ind="2" pos="PRP">I</word>
  <word ind="3" pos="VB">compare</word>
  <word ind="4" pos="PRP">you</word>
  <word ind="5" pos="TO">to</word>
  <word ind="6" pos="DT">a</word>
  <word ind="7" pos="NN">summer</word>
  <word ind="8" pos="POS">'s</word>
  <word ind="9" pos="NN">day</word>
  <word ind="10" pos=".">?</word>
</words>
= <dependencies>
= <dep type="aux">
  <governor idx="3">compare</governor>
  <dependent idx="1">Shall</dependent>
</dep>
= <dep type="nsubj">
  <governor idx="3">compare</governor>
```

```

    <dependent idx="2">I</dependent>
  </dep>
= <dep type="dobj">
  <governor idx="3">compare</governor>
  <dependent idx="4">you</dependent>
</dep>
= <dep type="prep">
  <governor idx="3">compare</governor>
  <dependent idx="5">to</dependent>
</dep>
= <dep type="det">
  <governor idx="7">summer</governor>
  <dependent idx="6">a</dependent>
</dep>
= <dep type="poss">
  <governor idx="9">day</governor>
  <dependent idx="7">summer</dependent>
</dep>
= <dep type="possessive">
  <governor idx="7">summer</governor>
  <dependent idx="8">'s</dependent>
</dep>
= <dep type="pobj">
  <governor idx="5">to</governor>
  <dependent idx="9">day</dependent>
</dep>
</dependencies>
= <nodes>
  <node id="1" parent="0" label="SQ" />
  <node id="2" parent="1" word="1" />
  <node id="3" parent="1" label="NP" />
  <node id="4" parent="3" word="2" />
  <node id="5" parent="1" label="VP" />
  <node id="6" parent="5" word="3" />
  <node id="7" parent="5" label="NP" />
  <node id="8" parent="7" word="4" />
  <node id="9" parent="5" label="PP" />
  <node id="10" parent="9" word="5" />
  <node id="11" parent="9" label="NP" />
  <node id="12" parent="11" label="NP" />
  <node id="13" parent="12" word="6" />
  <node id="14" parent="12" word="7" />
  <node id="15" parent="12" word="8" />
  <node id="16" parent="11" word="9" />
  <node id="17" parent="1" word="10" />
</nodes>
</Sentence>
</Document>

```

E5.2 Scansion output

=====TEXT ANALYSIS=====

=====Original Text=====

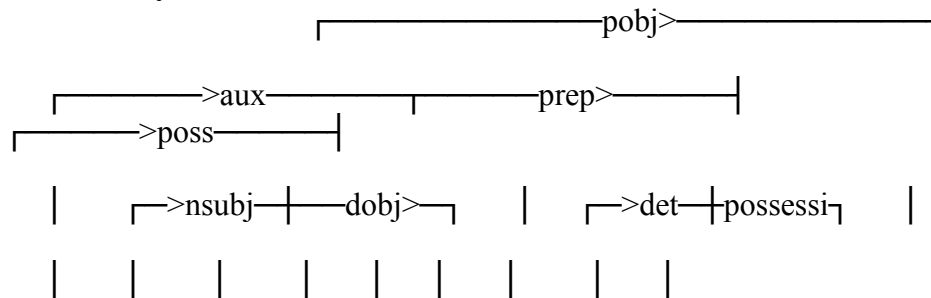
Shall I compare thee to a summer's day?

=====Revised Text=====

Shall I compare you to a summer's day?

=====SYNTACTIC ANALYSIS=====

=====Analysis=====



 Shall/VBD I/PRP compare/VB you/PRP to/TO a/DT summer/NN 's/POS day/NN ?. '

(SQ Shall

(NP I)

(VP compare

(NP you)

(PP to

(NP

(NP a summer 's) day))) ?)

=====PHONOLOGICAL ANALYSIS=====

=====Phonological data=====

[day*] [to a summer* 's] [Shall I compare* thee]

{[Shall I compare* thee]*}{[to a summer* 's] [day*]*}

<{[Shall I compare* thee]*} {[to a summer* 's] [day*]*}>

=====Speed of Phonological Phrase Processing=====
speed:31ms

=====Speed of Syllable Assignments=====
speed:485ms

=====STRESS ANALYSIS=====

=====Stress Patterns=====

Shall(0) I(0) compare(0) thee(0) to(0) a(0) summer(1)'s(-1) day(2)?/n/r shall(1) I(1)
Com(0)pare(2) thee(1) To(1) A(1) Sum(2)mer(0) 's(-1) Day(2)
nmwsnmnswm

=====Speed of Stress Assignment=====
speed:50ms

=====SCANSION=====

===== relative stresses + phonological data=====
<{[nm/ws\n]}mn/sw\]m]}>

=====line=====

Phonological:<{[nm/ws\n]}mn/sw\]m]}>
Stresses:xxwsxxxsws

=====HARTMAN SCANSION=====

Chosen Method: Maximise the Normal

=====Corral the Weird

xxwsxxxsws
xx/ws/wx/xs/ws/
RemainingSyllables:2
Start:
End:

=====Maximise Normal

xxwsxxxsws
xx/ws/wx/xs/ws/
MaxSyllableCount:8
StartsAt:0
EndsAt:7

=====PHONOLOGICAL SCANSION=====
All:<{[nm/ws\n]}mn/sw\]m]}>

KeyStresses:<{[xx/ws\n]}xx/sw\[m]}>
 Meter:< iambic pentameter
 Summary:(0,3)
 Iamb(R):7, 1IU, 1PP, 1PW Dactyl(R):5, 1IU, 1PW Anapaest(R):2, 1PW
 Iamb(F):6, 1IU, 1CP, 1PW Dactyl(F):5, 1IU, 1PW Anapaest(F):2, 1PW
 KeyRevised:<{[xx/ws\n]}xx/sw\[m]}>
 Scansion:xx/ws/nx/xs/wm
 Certainty:87.5%

=====Speed of Scansion=====
 speed:100ms

=====SPEECH XML=====

SSML:

<?xml version='1.0'?><speak version='1.0'
 xmlns='http://www.w3.org/2001/10/synthesis'
 xmlns:xsi='http://www.w3.org/2001/XMLSchema-instance'
 xsi:schemaLocation='http://www.w3.org/2001/10/synthesis
 http://www.w3.org/TR/speech-synthesis/synthesis.xsd' xml:lang='en-US'>

<pron sym="shall 1"/>
 <pitch absmiddle="5"><pron sym="I 1"/></pitch>
 <pron sym="Com"/>
 <pitch absmiddle="5"><pron sym="pare 2"/></pitch>
 <pron sym="thee 1"/>
 <pitch absmiddle="5"><pron sym="To 1"/></pitch>
 <pron sym="A 1"/>
 <pitch absmiddle="5"><pron sym="Sum 2"/></pitch>
 <pron sym="mer's 1"/>
 <pitch absmiddle="5"><pron sym="Day 2"/></pitch>
 </speak>